PRAKTIKUM SISTEM OPERASI

MODUL 10

SYSCALL

Oleh:

Zhafir Zaidan Avail 2311104059

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

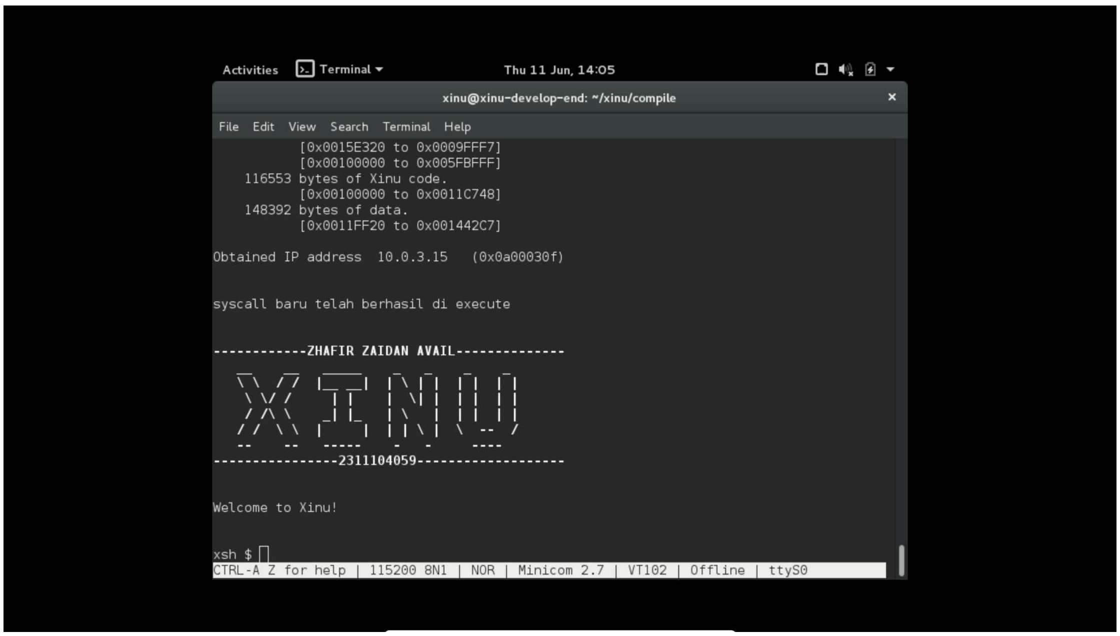
UNIVERSITAS TELKOM

2026

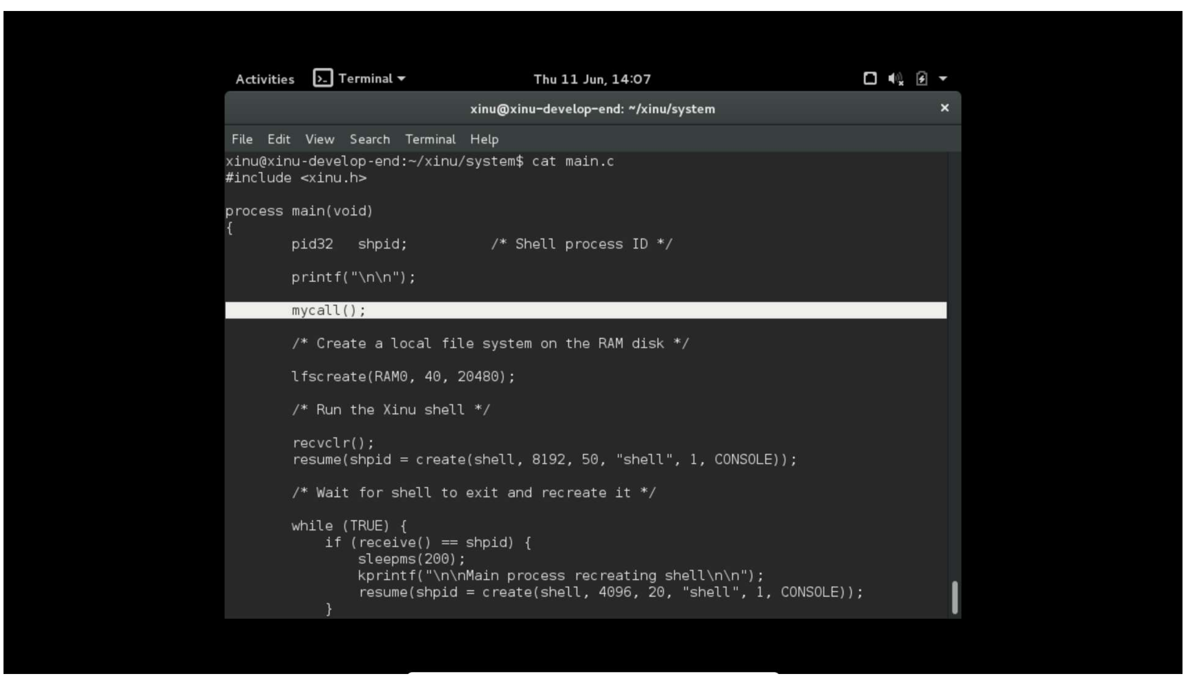
JURNAL/TP

1. **Soal 1**

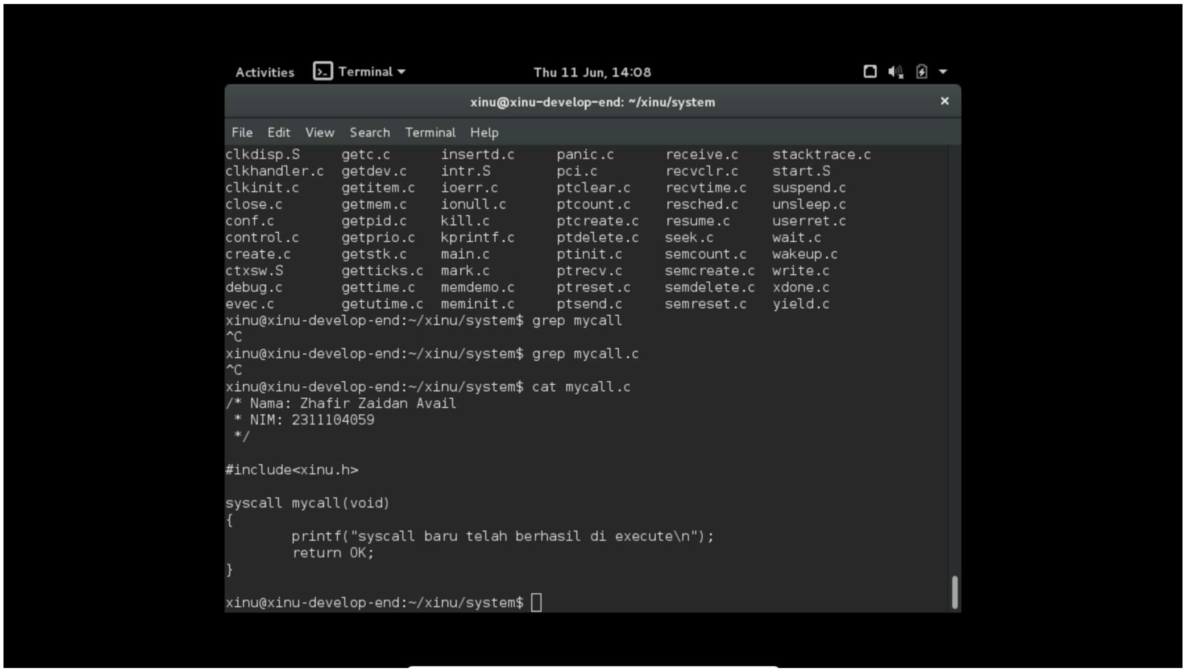
   [50 Poin] Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan Screenshot kode dan hasil run)



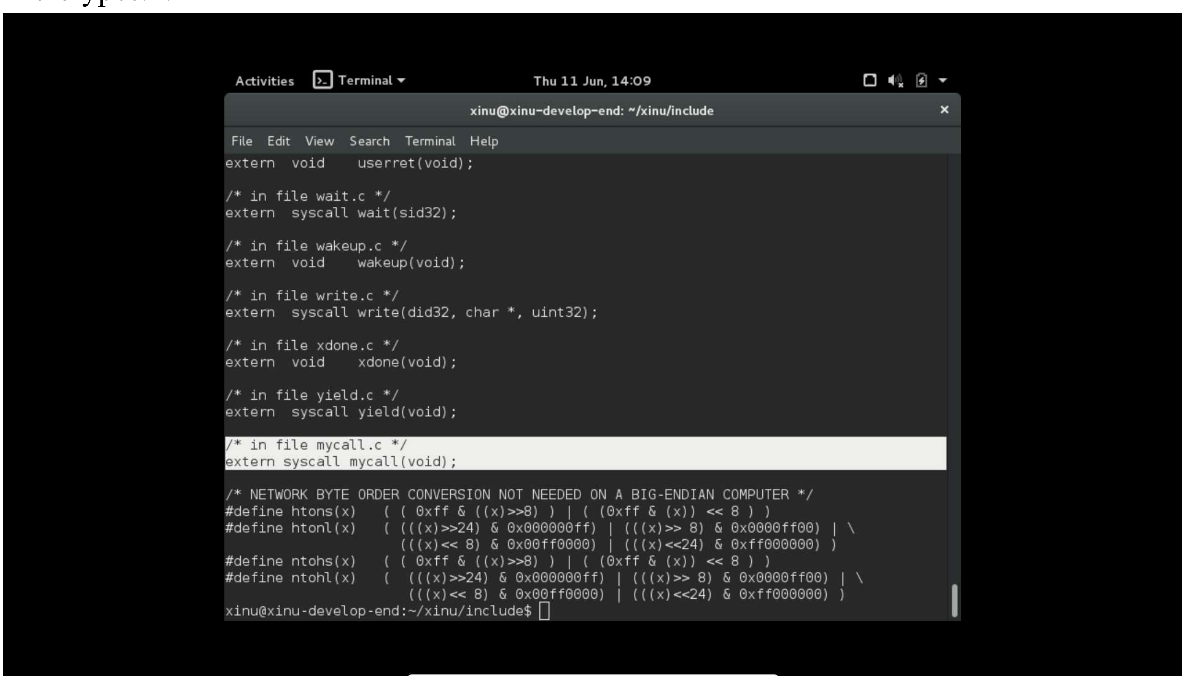
Code main.c:



Code mycall:



Prototypes.h:



2. **Soal 2**

   [25 Poin] Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input

   - Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
   - Pastikan prioritas adalah bilangan yang positif

   Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

   - make clean
   - make

Code pada chprio.c:

```
xinu@xinu-develop-end: ~/xinu/system
File  Edit  View  Search  Terminal  Help
          pri16         newprio         /* New priority                 */
        )
{
        intmask mask;                   /* Saved interrupt mask         */
        struct  procent *prptr;         /* Ptr to process's table entry */
        pri16   oldprio;                /* Priority to return           */

        mask = disable();
        if ((pid < 0) || (pid >= NPROC)) {
                restore(mask);
                return SYSERR;
        }

        if (newprio <= 0) {
                restore(mask);
                return SYSERR;
        }

        if (isbadpid(pid)) {
                restore(mask);
                return SYSERR;
        }
        prptr = &proctab[pid];
        oldprio = prptr->prprio;
        prptr->prprio = newprio;
        restore(mask);
        return 0xffff & oldprio;
}
```

Hasil makefile:

```
xinu@xinu-develop-end: ~/xinu/compile
File  Edit  View  Search  Terminal  Help
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/xsh_udpserver.o ../shell/xsh_udpserver.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/clkdisp.o ../system/clkdisp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$
```

3. **Soal 3**

Lakukan hal-hal berikut ini

- Edit xsh_uptime.c
  Tambahkan kode berikut

```
/* Check for valid number of arguments */

if (nargs > 1) {
        fprintf(stderr, "%s: too many arguments\n", args[0]);
        fprintf(stderr, "Try '%s --help' for more information\n",
                        args[0]);
        return 1;
}

// tambahan code
    chprio(5,33)   // memanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5
```

- Compile source code tersebut dengan perintah
  - make clean
  - make
- Jalankan perintah ps
  - `xsh $ ps`
  - perhatikan prioritas proses dengan id = 5
- Jalankan uptime
  - `xsh $ uptime`
  - Perhatikan hasil perintah tersebut
- Jalankan ps
  - `xsh $ ps`
  - perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

[25 Poin] Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0: Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c
- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”
- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)

tambahan di xhs_uptime.c: